

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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EXPERIMENT 4: VACUUM – CLEANER PROBLEM

Aim

The aim of this experiment is to implement a reflex-based Vacuum Cleaner agent that senses its environment and cleans a two-room grid space (Room A and Room B) by removing dirt.

Objective

To implement a simple reflex agent that perceives its current location and status to make immediate, rule-based cleaning and movement decisions without requiring long-term memory.

Algorithm

Step 1: Define the initial state of the environment, including the starting location of the vacuum cleaner (Room A or Room B) and the status of both rooms (Clean or Dirty).

Step 2: Read the current sensor input of the vacuum cleaner to determine its location and whether the current room is Clean or Dirty.

Step 3: If the status of the current room is Dirty, perform the Suck action to clean the room, and update its status to Clean.

Step 4: If the current room is already Clean, check the agent's location to decide the next move:

- If the agent is in Room A, execute the Move Right action to switch to Room B.
- If the agent is in Room B, execute the Move Left action to switch to Room A.

Step 5: Move to the adjacent room and repeat the sensor check to evaluate its cleanliness.

Step 6: If the room is Dirty, clean it using the Suck action.

Step 7: Once both rooms are confirmed to be Clean, stop the program and display the total performance score or step count.

Source Code

```
def reflex_vacuum_agent(location, status_a, status_b):  
    state = {'A': status_a, 'B': status_b}
```

```

current_loc = location

path = []

print(f"Initial State -> Location: {current_loc}, Room A: {'Dirty' if status_a else 'Clean'},
Room B: {'Dirty' if status_b else 'Clean'}\n")

# Run loop until both rooms are clean
while state['A'] == 1 or state['B'] == 1:

    path.append(f"({current_loc}, {'Dirty' if state[current_loc] else 'Clean'})")

    if state[current_loc] == 1:

        print(f"Location {current_loc} is Dirty. Action: SUCK")

        state[current_loc] = 0 # Clean the room

    else:

        if current_loc == 'A':

            print("Location A is Clean. Action: MOVE RIGHT")

            current_loc = 'B'

        else:

            print("Location B is Clean. Action: MOVE LEFT")

            current_loc = 'A'

# Append final clean state
path.append(f"({current_loc}, Clean)")

print(f"\nBoth rooms are clean! Final state path:")

print(" -> ".join(path))

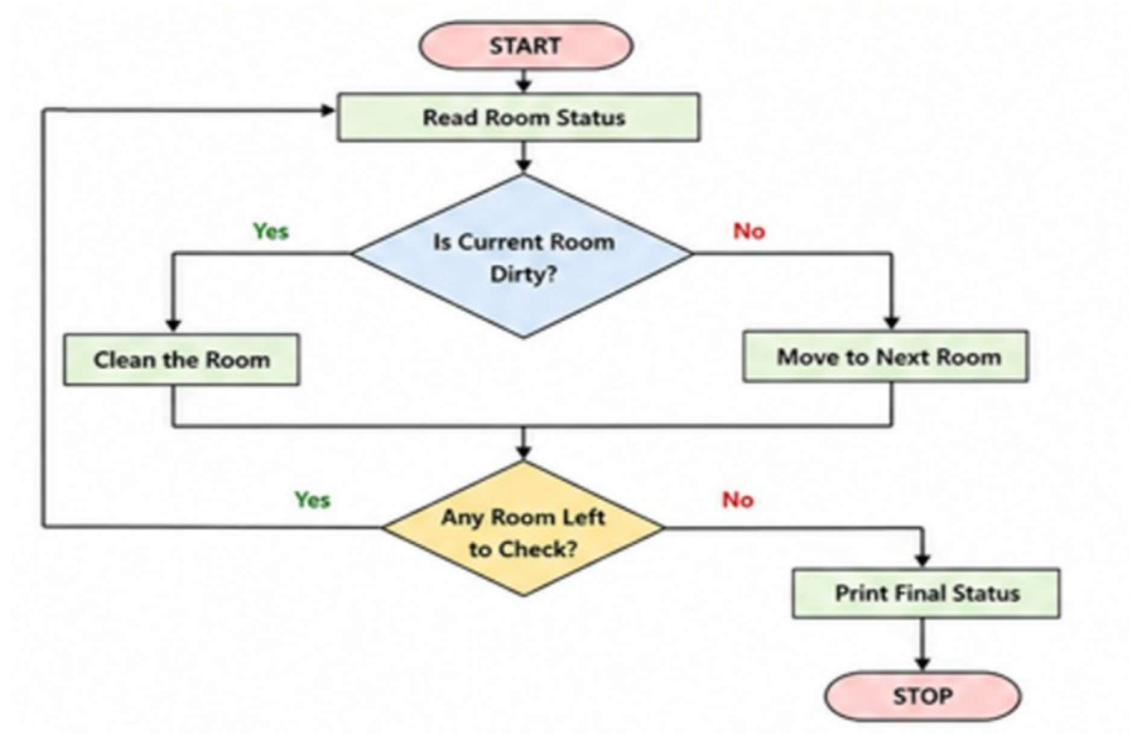
# --- Run the Simulation ---

# Parameters: (Initial Location, Status A, Status B) | 1 = Dirty, 0 = Clean

```

reflex_vacuum_agent(location='A', status_a=1, status_b=1)

Flowchart



Output

```
>>>
==== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/EXPl.py ====
Initial State -> Location: A, Room A: Dirty, Room B: Dirty

Location A is Dirty. Action: SUCK
Location A is Clean. Action: MOVE RIGHT
Location B is Dirty. Action: SUCK

Both rooms are clean! Final state path:
(A, Dirty) -> (A, Clean) -> (B, Dirty) -> (B, Clean)
>>>
```

Result

The Vacuum Cleaner World environment was successfully simulated, and the reflex agent efficiently cleaned all dirty rooms using optimal movement patterns.